

Muon sites in PbF_2 and YF_3 : decohering environments and the role of anion Frenkel defects

Supplemental information

J. M. Wilkinson,^{1,*} F. L. Pratt,² T. Lancaster,³ P. J. Baker,² and S. J. Blundell^{1,†}

¹*Clarendon Laboratory, University of Oxford Department of Physics,
Parks Road, Oxford, OX1 3PJ, United Kingdom*

²*ISIS Facility, STFC Rutherford Appleton Laboratory, Didcot OX11 0QX, United Kingdom*

³*Centre for Materials Physics, Durham University, Durham DH1 3LE, United Kingdom*

(Dated: October 29, 2021)

I. μSR EXPERIMENTS

In the muon experiment, a beam of spin-polarized muons were incident on a sample, and the number of positrons detected in both the forwards and backwards detectors, $N_F(t)$ and $N_B(t)$ respectively, was measured [1]. The muon asymmetry was calculated as

$$A(t) = \frac{N_B(t) - \alpha N_F(t)}{N_B(t) + \alpha N_F(t)}, \quad (1)$$

where the parameter α takes into account systematic differences between the readings of both sets of detectors. Our experiments were performed using the MuSR spectrometer at the ISIS Facility, Rutherford Appleton Laboratory, UK. Polycrystalline samples of $\alpha\text{-PbF}_2$ and YF_3 were wrapped in sheet of 25 μm silver foil, and were placed in Variox cryostats, kept at a temperature of 10 K, and at zero field. The Earth's magnetic field was compensated to better than 50 μT using active field compensation. Additional 20 million event runs were also taken on PbF_2 with applied longitudinal fields of 5, 10 and 20 Gauss, also at 10 K. The mean muon lifetime is 2.2 μs , but data can be obtained out to at least ten times this value at ISIS if collected for several hours. Around 360 million and 274 million muon decay events were detected for $\alpha\text{-PbF}_2$ and YF_3 respectively. These statistics are much higher than are normally taken for muon data, highlighting the importance of high data rate acquisition of time-differential data at pulsed sources. High statistics are needed to resolve long-time features which are important in revealing the details of the more distant couplings. As explained in the main paper, our fits do not use any phenomenological relaxation function multiplying the coherent oscillations. Because only a relatively small number of neighbouring nuclear spins are included in our calculations (ten for these cases), the data and the fits will always diverge at very long times, but by taking high statistics data, and including as many nearest neighbours as the computations allow, we are able to achieve good agreement to fairly late times.

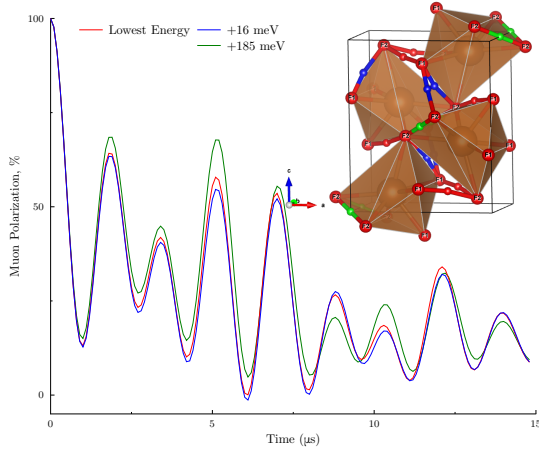
II. DFT+ μ CALCULATIONS

The ab initio calculations were performed with the QUANTUM ESPRESSO package [2]. The calculations were performed in a supercell containing $2 \times 2 \times 2$ conventional unit cells, using a $3 \times 3 \times 3$ Monkhorst-Pack grid, and ultrasoft pseudopotentials to model the electronic wavefunctions of the atoms. For the states considered here, the +1 and -1 charge states of the bare muon (μ^+) and muonium anion (Mu^-) were determined by the charge of the supercell. A muon was placed in several randomly chosen low-symmetry sites and all ions were allowed to relax until the forces on all ions had fallen below a convergence threshold of 51 meV/ $\text{\AA}^{-1}$, and the total energy had fallen below a convergence threshold of 2.7 meV.

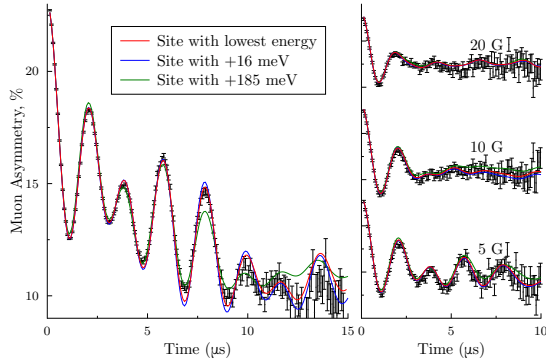
For $\alpha\text{-PbF}_2$, our DFT analysis predicts three possible low-energy sites. The two with the lowest energies are between crystallographically inequivalent fluoride anions (one is only 16 meV higher than the other) and the site with the third lowest energy (185 meV above the lowest energy site) places the muon between two fluoride ions which occupy the same Wyckoff position [?]. The predicted muon polarization from all these sites is plotted in Supplemental Figure 1, where it can be seen that there is only a small difference between the two sites with the lowest energy, but the site with energy above this shows a substantially different time-dependent polarization, and can therefore be ruled out (note that the polarization plotted in this figure is before any small adjustment of the $\mu\text{-F}$ distances which occur when these polarizations are fitted to the data). These three distinct muon sites were used to fit the data, as shown in Figure 2, where it can be seen that the lowest energy (red line) provides the best fits, although the difference between this and the site with energy 16 meV above this is marginal. However, the muon-induced distortions can be derived from these fits, and compared to the corresponding DFT values, see Supplemental Table I. This table suggests that the lowest energy site is indeed the correct muon stopping site, as the values of most of the parameters are very close to that calculated with DFT, especially when considering the next-nearest neighbours. However, more statistics are required to be able to conclusively distinguish between these sites.

* john.wilkinson@physics.ox.ac.uk

† stephen.blundell@physics.ox.ac.uk



Supplemental Fig. 1. The polarization of the three muon sites in α -PbF₂, predicted by DFT+ μ to have the lowest total energy. The inset shows the muon sites with the three lowest energies as red, blue and green spheres (ignoring muon-induced distortions), and the time dependence of muon polarization expected from these three sites is shown in the main frame.

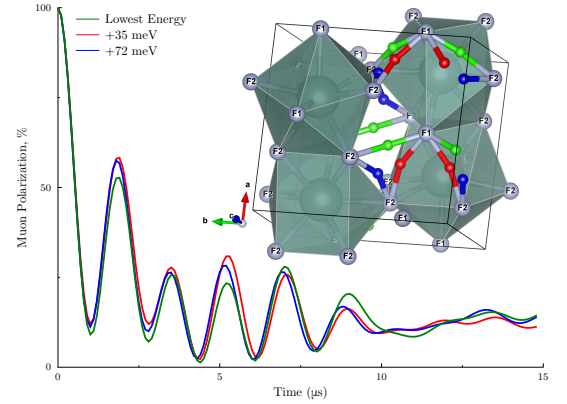


Supplemental Fig. 2. Fitting the data of PbF₂ with the muon sites as described in the text. The fit which was deemed the best has the muon positioned at the site calculated by DFT to have the lowest energy.

For YF₃, the DFT+ μ calculation resulted in three sites which had the lowest energy: two between crystallographically inequivalent fluorines, and one between two fluorines in equivalent sites, as shown in the inset in Supplemental Figure 3. The muon polarization for these three states are also plotted in the figure, and the site which was determined to best fit the data was the site with the lowest overall energy. This site was confirmed by the perturbations of the surrounding atoms being very close to the values predicted by DFT, as described in the main text.

	Parameter	Experimental	DFT+ μ	Difference
Lowest Energy	r_{nn1} (Å)	1.1419(4)	1.0967	+0.0452
	r_{nn2} (Å)	1.2601(7)	1.2057	+0.0544
	r_{nnn1} (Å)	2.68(3)	2.89	-0.21
	r_{nnn2} (Å)	3.01(7)	3.27	-0.26
	ζ_4	0.834(5)	0.838	-0.004
+16 meV	r_{nn1} (Å)	1.1410(5)	1.0965	+0.0445
	r_{nn2} (Å)	1.2580(9)	1.2090	+0.0490
	r_{nnn1} (Å)	2.54(3)	2.71	-0.17
	r_{nnn2} (Å)	3.54(1)	2.94	+0.6
	ζ_4	0.851(7)	0.835	+0.016

Supplemental Table I. Muon-induced structural distortions in PbF₂, comparing the values obtained with the fit to experimental data with the values obtained using DFT+ μ , using both the lowest-energy site and the site with energy 16 meV above this.



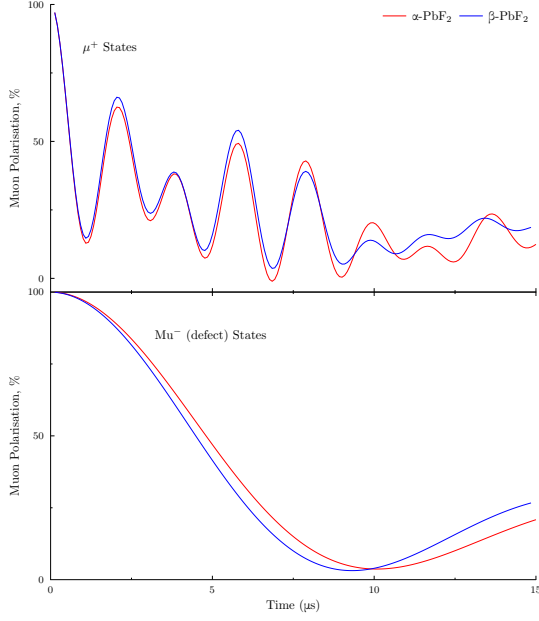
Supplemental Fig. 3. The polarization of the muon sites predicted by DFT+ μ to have the lowest total energy. The inset shows the muon sites with the lowest energy (green), 35 meV above this (red), and 72 meV above the lowest energy site (blue) (the distortions induced by the muon are not shown). The graph in the main frame corresponds to the expected muon polarization from these three sites.

III. CALCULATION OF THE PbF₂ POLARISATION

The muon polarisation of both phases of PbF₂ (after the muon perturbations were fitted to the data) are shown in Supplemental Figure 4.

A. Mu⁻ states in defect-free α -PbF₂

DFT calculations were also undertaken on Mu⁻ states in a defect-free α -PbF₂ crystal, to find the likely stopping site and hence calculate how such a state would be visible in the muon polarization. These results are shown in Supplemental Figure 5, where it can be seen that the simulation for the Mu⁻ state in the defect-free lattice is



Supplemental Fig. 4. PbF₂ Polarisation: The top graph shows the “bare” muon (μ^+) polarisation for both phases of PbF₂, and the bottom graph shows this for the Mu^- defect states as explained in the main text.

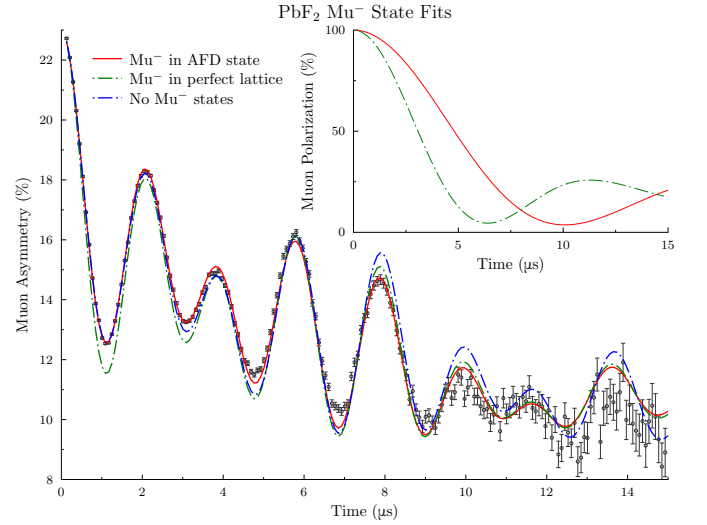
not only vastly different to that of the state in the AFD, but also fails to match the data in a satisfactory way. Hence, the existence of such Mu^- states in the pristine lattice in this compound can be ruled out.

B. $\alpha\text{-PbF}_2$

$\alpha\text{-PbF}_2$ has space group $Pnma$, with Pb at $0.2428 \frac{3}{4} 0.5982$, and F at $0.0172 \frac{3}{4} 0.3520$ and $0.1415 \frac{3}{4} 0.9291$ [4]. The nuclei involved in calculating the bare muon polarization were originally at the fractional coordinates in Supplemental Table II. The muon was then added to the crystal at the site predicted by the DFT+ μ calculations, and the distance between the muon and the two nearest neighbours F_{nn1} and F_{nn2} , r_{nn1} and r_{nn2} respectively, were varied by the fitting algorithm, along with the distance between the muon and the two next-nearest neighbours F_{nnn1} and F_{nnn2} , with parameters r_{nnn1} and r_{nnn2} (the DFT+ μ calculation showed that the positions of the fluorines further than this did not change significantly). The parameter ζ_4 was also fitted by scaling the F- μ couplings of F_7 , F_8 , F_9 and F_{10} .

C. $\beta\text{-PbF}_2$

$\beta\text{-PbF}_2$ has a fluorite structure with space group $Fm\bar{3}m$, with Pb at 000 and F at $\frac{1}{4} \frac{1}{4} \frac{1}{4}$ [5]. The nuclei in-



Supplemental Fig. 5. PbF₂ Mu^- state fits: The main plot shows the fit of the muon asymmetry from the model including a Mu^- state in crystals both with an AFD and without an AFD (the “perfect lattice”), and without including any Mu^- states at all. Inset: the muon polarization exclusively from the two Mu^- states, i.e. without the μ^+ component.

Spin	Fractional Position	
μ	0.1879	$\frac{1}{2}$ 0.3906
F_{nn1}	0.0172	$\frac{3}{4}$ 0.3520
F_{nn2}	0.3586	$\frac{1}{4}$ 0.4291
F_{nnn1}	-0.0172	$\frac{1}{4}$ 0.6480
F_{nnn2}	0.5172	$\frac{3}{4}$ 0.1470
F_5	0.0172	$\frac{1}{4}$ 0.3520
F_6	0.3585	$\frac{5}{4}$ 0.4291
F_7	0.6415	$\frac{3}{4}$ 0.5709
F_8	-0.1415	$\frac{1}{4}$ 0.0709
F_9	0.1415	$\frac{3}{4}$ -0.0709
F_{10}	-0.0172	$\frac{5}{4}$ 0.6480

Supplemental Table II. The original positions of the nuclei in $\alpha\text{-PbF}_2$, before muon implantation.

involved in calculating the bare muon (μ^+) polarisation are tabulated in Supplemental Table III. The parameter r_{nn} was used to fit the nearest-neighbour fluorine distances to the muon, and the distances between the muon and the next-nearest neighbours F_{nnnx} were rescaled by the parameter ζ_{nnn} .

IV. CALCULATION OF THE YF_3 POLARISATION

YF_3 has space group $Pnma$, with Y at 000, and F at $0.1652 \ 0.0643 \ 0.3755$ and $0.5227 \ \frac{1}{4} \ 0.5910$ [3]. The nuclei involved in the calculation were originally at the frac-

Spin	Fractional Position		
μ	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{4}$
F_{nn1}	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}$
F_{nn2}	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{3}{4}$
F_{nnn1}	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
F_{nnn2}	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
F_{nnn3}	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{5}{4}$
F_{nnn4}	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{5}{4}$
F_{nnn5}	$\frac{1}{4}$	$-\frac{1}{4}$	$\frac{3}{4}$
F_{nnn6}	$\frac{3}{4}$	$-\frac{1}{4}$	$\frac{3}{4}$
F_{nnn7}	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
F_{nnn8}	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

Supplemental Table III. The original positions of the nuclei in β -PbF₂, before muon implantation.

tional coordinates in Supplemental Table IV. The muon perturbations were then estimated with DFT+ μ , and then fine-tuned with the fitting parameters r_x to adjust the muon–fluorine distances for the x nearest-neighbour nuclei, and ζ_6 to rescale the distance between F_{5-10} and the muon to account for all the nuclei in the crystal.

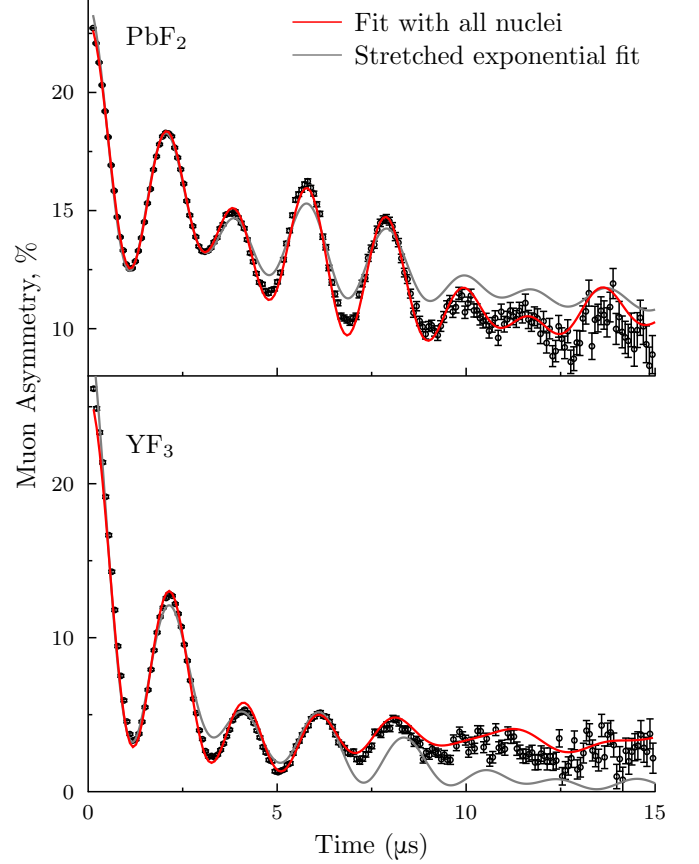
Spin	Fractional Position		
μ	0.5635	0.9233	0.3378
F_{nn1}	0.4733	0.7500	0.4090
F_{nn2}	0.6652	1.0643	0.1245
F_{nnn1}	0.8348	0.9357	0.6245
F_{nnn2}	0.3348	0.9357	-0.1245
F_5	0.5227	1.2500	0.5910
F_6	0.1652	1.0643	0.3755
F_7	0.3348	0.9357	-0.1245
F_8	0.9773	0.7500	0.0910
F_9	0.8348	0.9357	-0.3755
F_{10}	0.6652	0.4357	0.1245

Supplemental Table IV. Original positions of the nuclei in YF₃, before muon implantation.

V. COMPARING THE FITS WITH PREVIOUS APPROACHES

The previous approach to fitting F– μ –F data is to take the coherent “F– μ –F” function, and multiply it by a stretched exponential function to model the decohering

environment. Although this can sometimes produce visually acceptable fits, [6, 7], the stretched exponential function does not have any physical meaning in this context. In Supplemental Figure 6, we show the best fits we could obtain to our data using the stretched exponential function, in comparison to the fit obtained by the method



Supplemental Fig. 6. Fits to our data for PbF₂ (top graph) and YF₃ (bottom) with an F– μ –F function multiplied by a stretched exponential (red), and the fits considering all the nuclei in the crystal (gray).

described in the main text. Although χ^2 per d.o.f is not a particularly useful metric for high rate μ SR data, as the early time data is affected strongly by deadtime, our method gives values which are significantly better: For PbF₂, the fit using all nuclei gives a χ^2 per d.o.f of 3.99, but the stretched exponential fit gives 25.58. For YF₃, the all nuclei fit gives 16.84, but the stretched exponential fit gives 48.23.

[1] S. F. J. Cox, J. Phys. C: Solid State Phys. **20**, 3187 (1987).

[2] P. Giannozzi et al. J. Phys.: Condens. Matter **21**, 395502 (2009).

- [3] A. Zalkin and D. H. Templeton, J. Am. Chem. Soc. **75** 2453 (1953).
- [4] L. Ehm et al. J. Phys. Chem. Sol. **64** 919 (2003)
- [5] Y. Ito and K. Koto Solid State Ionics, **9-10** 527 (1983)
- [6] J. Brewer et al. Phys. Rev. B, **33** 7813 (1986)
- [7] D. R. Noakes et al. J. Phys. Chem. Solids **54** 785 (1993)